TEST CODE **000592**

FORM TP 96104

MAY/JUNE 1996

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 - General Proficiency

2 hours 40 minutes

05 JUNE 1996 (a.m.)

INSTRUCTIONS TO CANDIDATES

1. **Answer ALL questions in Section I, and any TWO in Section II.**
2. **Begin the answer for each question on a new page.**
3. **Full marks may not be awarded unless full working or explanation is shown with the answer.**
4. **Mathematical tables, formulae and graph paper are provided.**
5. **Silent electronic calculators may be used for this paper.**
6. **You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Copyright © 1995 Caribbean Examinations Council.
All rights reserved.

SECTION I

Answer ALL the questions in this section.

1. (a) Calculate the value of

$$\frac{2\frac{1}{3} - 1\frac{1}{2}}{1\frac{5}{6}}$$, expressing your answer as a fraction in lowest terms. **(2 marks)**

(b) Evaluate $\sqrt{1.073}$, giving your answer to three significant figures. **(2 marks)**

(c) The Peters' family consists of Mr Peters, his wife and two children. Mr Peters' monthly salary is $2,800.

Table showing income tax allowances for a year are calculated as follows:

Husband	$1000.
Wife	$ 600.
Each child	$ 200.

Table showing Tax Rates for a year

Chargeable Income	Rate of tax
For every dollar of the first $12 000.	5 cents
" " " " " next $ 8 000.	15 cents
" " " " " next $20 000.	35 cents

Calculate for Mr Peters,

(i) his salary for the year **(1 mark)**

(ii) his total allowances **(2 marks)**

(iii) his chargeable income **(1 mark)**

(iv) the amount he has to pay as income tax **(2 marks)**

(v) the percent of his salary paid as income tax. **(2 marks)**

Total 12 marks

2. (a) Given that $l = -2$, $m = 3$ and $n = 7$,

calculate the value of

$lm(m - n)$. **(2 marks)**

(b) If $\frac{p}{s} = \frac{q}{s} + r$, express s in terms of p, q and r. **(2 marks)**

GO ON TO THE NEXT PAGE

(c) Factorize completely:

(i) $8h^2 - 4h$

(ii) $4a^2 - 1$ **(3 marks)**

(d) Solve the equation

$$\frac{4x + 5}{4} - \frac{9 + 2x}{3} = 0.$$ **(3 marks)**

Total 10 marks

3. (a)

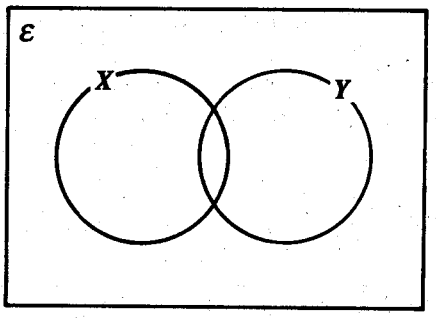


Figure 1

The Venn Diagram (Figure 1) shows two sets, X and Y, which are subsets of the universal set ε.

Copy the diagram and shade the region which represents $(X \cup Y) \cap X'$.

(1 mark)

(b)

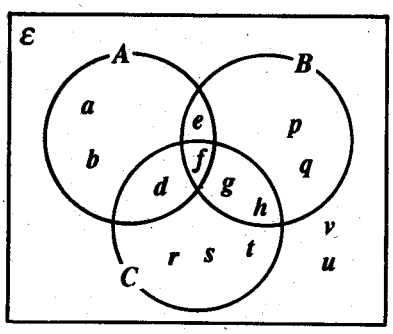


Figure 2

Figure 2 is a Venn Diagram showing the universal set ε and three subsets A, B and C. The lower case letters written in the diagram are members of the various subsets.

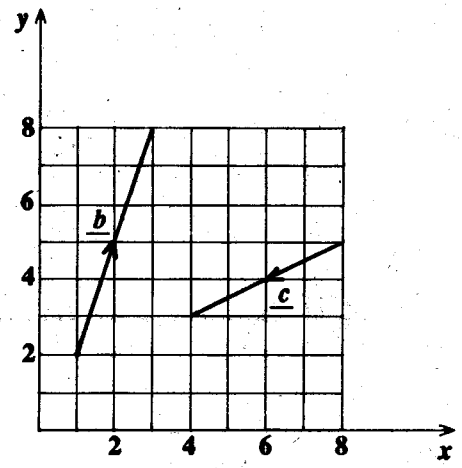
(i) List the members of $(A \cup B \cup C)'$. **(1 mark)**

(ii) List the members of $B \cap C$. **(1 mark)**

(iii) State the value of $n(B \cup C)$. **(1 mark)**

GO ON TO THE NEXT PAGE

(c)



The diagram above shows vector $\underline{b}$ and vector $\underline{c}$.

Express in the form $\begin{pmatrix} p \\ q \end{pmatrix}$

(i) $\underline{b}$ **(1 mark)**

(ii) $\underline{c}$ **(1 mark)**

(iii) $\underline{b} - \underline{c}$. **(2 marks)**

Calculate

(iv) $|\underline{b} - \underline{c}|$ **(2 marks)**

(v) the size of the angle between $\underline{b} - \underline{c}$ and the x-axis. **(2 marks)**

Total 12 marks

4. (a) (i) Construct triangle LMN with MN =10 cm, LM = 6.5 cm and angle LMN = 50°. **(2 marks)**

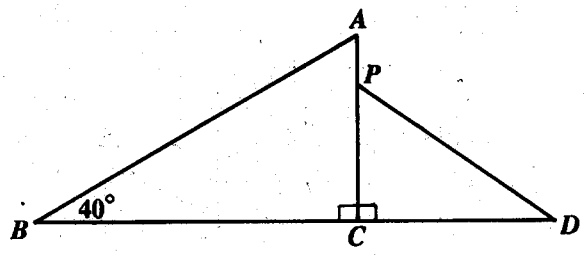
(ii) Measure and state the length of LN. **(1 mark)**

(iii) Measure and state the size of angle LNM. **(1 mark)**

(iv) Using ruler and compasses ONLY, construct LX so that LX is perpendicular to MN and meets MN in X. **(2 marks)**

GO ON TO THE NEXT PAGE

(b)



The diagram above, **not drawn to scale**, shows ABC and PCD are right angled triangles. Angle $ABC = 40°$, $AB = 10$ cm, $PD = 8$ cm and $BD = 15$ cm.

Calculate, giving your answers correct to 1 decimal place

(i) the length of BC in centimetres **(2 marks)**

(ii) the size of angle PDC in degrees. **(3 marks)**

Total 11 marks

5. (a) If $f(x) = 2x - 1$ and $g(x) = \frac{1}{2}(x + 2)$,

calculate

(i) $f(3)$ **(1 mark)**

(ii) $g^{-1}(x)$ **(2 marks)**

(iii) $gf(3)$ **(2 marks)**

(b) Write your answers to this question on the answer sheet provided.

The diagram drawn on the answer sheet shows triangle ABC.

T is a translation vector represented by $\begin{pmatrix} -6 \\ 0 \end{pmatrix}$.

R is 180° rotation about the point (5, 3).

(i) Write down the coordinates of A, B and C. **(2 marks)**

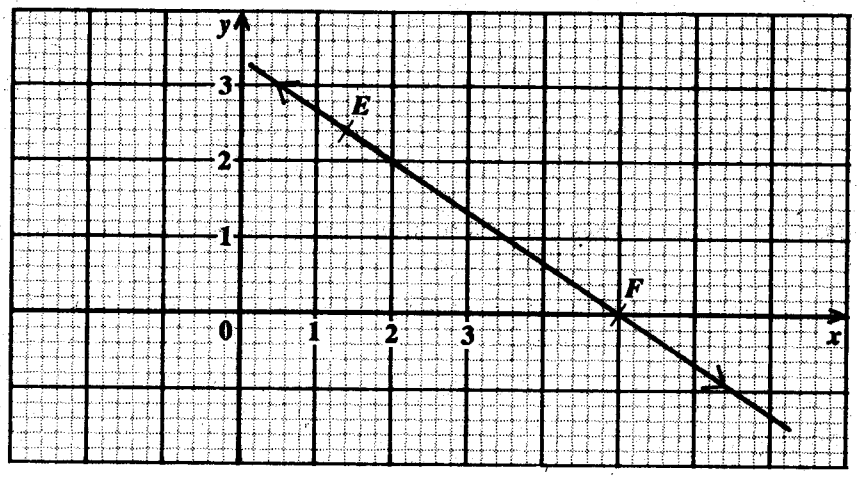
(ii) Draw and label triangle $A'B'C'$ which is the image of triangle ABC under T. **(2 marks)**

(iii) Draw and label triangle $A''B''C''$ which is the image of triangle ABC under R. **(2 marks)**

Total 11 marks

GO ON TO THE NEXT PAGE

6. (a)



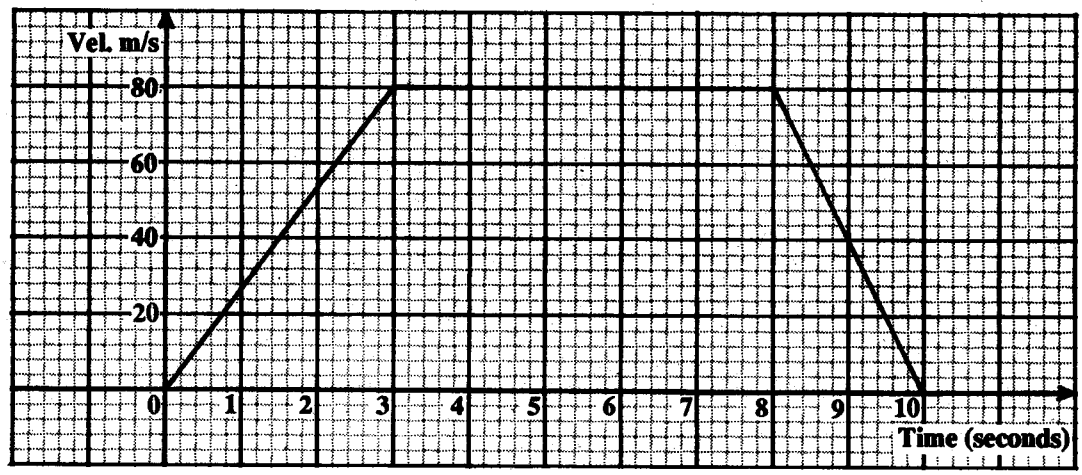
The graph above shows a straight line EF intersecting the x-axis at F.

(i) State the coordinates of F. **(1 mark)**

(ii) Calculate the gradient of EF. **(2 marks)**

(iii) Determine the equation of EF. **(2 marks)**

(b)



The velocity-time graph shown above represents the journey of a motor cyclist, between two points.

Using the graph, determine, by stating the units used,

(i) the length of time during which the cyclist accelerated **(1 mark)**

(ii) the acceleration **(3 marks)**

(iii) the length of time during which the velocity was constant **(1 mark)**

(iv) the deceleration. **(2 marks)**

Total 12 marks

GO ON TO THE NEXT PAGE

7. (a)

Year	1990	1991	1992	1993	1994	1995
Cost (dollars)	10 000	30 000	36 000	30 000	34 000	50 000

The table above shows the cost of a plot of land over a period of time.

(i) On graph paper, using a scale of 2 cm to represent 1 year on the horizontal axis and 1 cm to represent $10,000 on the vertical axis, draw a **line graph** to represent the data. **(4 marks)**

From your graph:

(ii) Determine the period during which the cost of the plot of land showed the greatest rate of increase. **(1 mark)**

(iii) State the year during which the cost of the plot of land decreased. **(1 mark)**

(iv) Estimate the cost of the plot of land in the year 1996. **(1 mark)**

(v) State one reason for your estimate in (iv) above. **(1 mark)**

(b) A firm manufacturing computers found that on the average 65 out of 80 of its computers passed the required standard set for quality control.

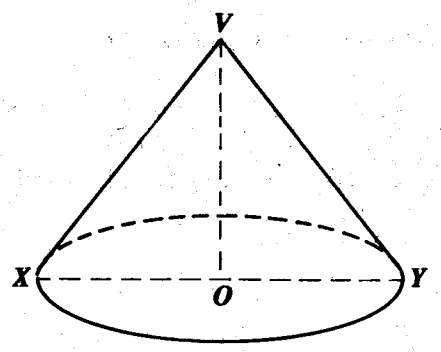
Calculate

(i) the probability that a computer selected at random would be defective **(2 marks)**

(ii) the number of computers the firm would expect to be defective, if it manufactured 336 computers. **(2 marks)**

Total 12 marks

8. (a)



The diagram VXY above, **not drawn to scale**, represents a right open-ended paper cone. The slant height VX is 14 cm, the base diameter XY is 21 cm and VO is the height of the cone.

[Take π to be $\frac{22}{7}$].

(i) Calculate, in cm, the circumference of the base of the cone. **(1 mark)**

(ii) Show that the height of the cone is approximately 9.3 cm. **(2 marks)**

(iii) Calculate the volume, in cm^3, of the cone. **(2 marks)**

(b) The cone is cut along a straight line drawn from Y to V.

The paper is flattened out to form a plane figure.

(i) Draw a diagram of the plane figure, showing the position of V. **(1 mark)**

(ii) Write the length of each side on your diagram. **(2 marks)**

(iii) Calculate the size of the angle, in degrees, at V. **(2 marks)**

Total 10 marks

SECTION II

Answer TWO questions in this section.

RELATIONS AND FUNCTIONS

9. (a) Given that $f(x) = x^3$, copy and complete the table below:

x	-3	-2	-1	0	1	2	3
$f(x)$	-27			0		8	27

(2 marks)

(b) Using 2 cm to represent 1 unit on the x-axis and 1 cm to represent 5 units on the y-axis, draw the graph of $f(x) = x^3$ for $-3 \leq x \leq 3$. **(5 marks)**

(c) On the graph of $f(x) = x^3$:

(i) Draw a tangent to the curve at the point $x = 2$. **(2 marks)**

(ii) Estimate the gradient of $f(x)$ at $x = 2$. **(2 marks)**

(iii) Draw the graph of $g(x) = -6x - 20$. **(2 marks)**

(d) Using the graphs, obtain the value of x for which $f(x) = g(x)$. **(2 marks)**

Total 15 marks

10. A florist uses x lilies and y roses to make bouquets.

The total number of flowers in a bouquet must be less than or equal to 45.

(i) Write an inequality in x and y to represent this condition. **(1 mark)**

(ii) There must be at least 10 lilies in the bouquet.
Write an inequality to represent this condition. **(1 mark)**

(iii) The number of lilies is less than or equal to twice the number of roses.
Write an inequality in x and y to represent this condition. **(2 marks)**

(iv) Using a scale of 1 cm to 5 units on each axis, graph the inequalities in (i), (ii) and (iii) above. **(5 marks)**

(v) Shade ONLY the region which satisfies the three inequalities in (i), (ii) and (iii) above. **(2 marks)**

(vi) The profit on a lily is \$1, and the profit on a rose is \$2.
Write an expression for the total profit made on a bouquet. **(1 mark)**

(vii) Use the graph to determine the maximum profit that could be made on a bouquet. **(3 marks)**

Total 15 marks

GO ON TO THE NEXT PAGE

TRIGONOMETRY

11. (a) Two ships leave a Port O.
One ship travels on a bearing of 340° to a point P which is 50 km from O.
The other ship travels on a bearing of 060° to a point Q, 85 km from O.

(i) Draw a diagram to represent the position of the port and the two ships.
On your diagram, carefully label North, the given angles and the distances travelled. **(4 marks)**

(ii) Calculate the distance PQ in km. **(3 marks)**

(iii) Determine the bearing of P from Q. **(4 marks)**

(b) Given that $\tan\theta = \frac{2}{\sqrt{3}}$ and θ is an acute angle,

show that $\sin\theta\cos\theta = \frac{2\sqrt{3}}{7}$ **(4 marks)**

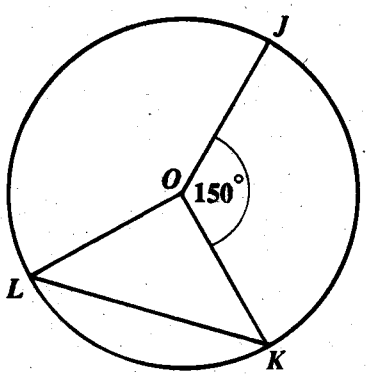
Total 15 marks

12. (a) Convert

(i) 160° to radians

(ii) $\frac{2\pi}{3}$ radians to degrees. **(2 marks)**

(b)



The diagram above, **not drawn to scale**, shows a circle of radius 10 cm.
O is the centre of the circle.
J, K and L are points on the circumference of the circle.
Angle $JOK = 150°$.
Take π to be 3.14.

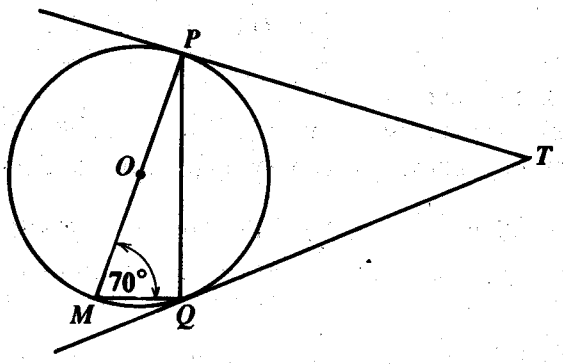
(i) Calculate the length in cm of the minor arc JK. **(2 marks)**

(ii) Show that the area of the minor sector OJK is $\frac{125\pi}{3}$ cm^2. **(2 marks)**

(iii) If the area of the triangle LOK is 43.3 cm^2, calculate, in degrees, the size of the angle LOK. **(3 marks)**

GO ON TO THE NEXT PAGE

(c)



The diagram above, **not drawn to scale**, shows TP and TQ as tangents to a circle from a point T. The circle has centre O and the angle $OMQ = 70°$.

Calculate, giving reasons for your answer, the measure of

(i) angle POQ **(2 marks)**

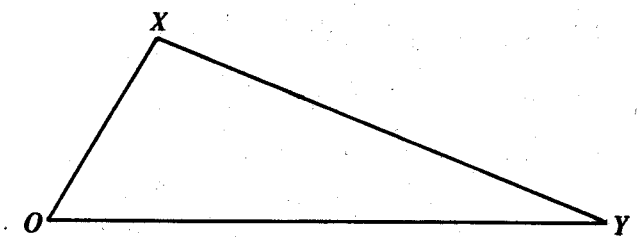
(ii) angle MPQ **(2 marks)**

(iii) angle PTQ. **(2 marks)**

Total 15 marks

VECTORS AND MATRICES

13.



In the triangle OXY, $\overrightarrow{OX} = \underline{x}$ and $\overrightarrow{OY} = \underline{y}$.

M is the mid-point of XY.

$\overrightarrow{OL} = \frac{3}{4}\overrightarrow{OX}$ and $\overrightarrow{ON} = \frac{3}{2}\overrightarrow{OY}$.

(i) Copy the triangle OXY and indicate on the triangle, the positions of L, M and N. **(2 marks)**

(ii) Express in terms of $\underline{x}$ and $\underline{y}$

$\overrightarrow{XY}$, $\overrightarrow{LM}$ and $\overrightarrow{MN}$. **(9 marks)**

Using a vector method,

(iii) Prove L, M and N are collinear. **(2 marks)**

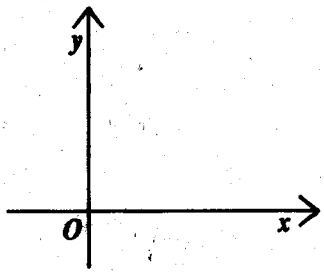
(iv) State the value of the ratio

$LM : MN$. **(2 marks)**

Total 15 marks

GO ON TO THE NEXT PAGE

14.



A plane is defined by two perpendicular coordinate axes, Ox and Oy.

(i) Write down a 2 x 2 matrix which represents R, a reflection in the line $y = x$. **(1 mark)**

(ii) Write down a 2 x 2 matrix which represents S, an anticlockwise rotation through 180° about the origin. **(1 mark)**

(iii) Describe, in words, the transformation Q which is represented by the vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. **(1 mark)**

(iv) Determine the image of the point (3, 4) under the following transformation

RS **(3 marks)**
SQ. **(2 marks)**

(v) Describe, in words, RS as a single geometrical transformation. **(3 marks)**

(vi) Calculate the values of a and b such that the point (a, b) is an invariant point under the transformation SQ. **(4 marks)**

Total 15 marks

END OF TEST

TEST CODE **000592**

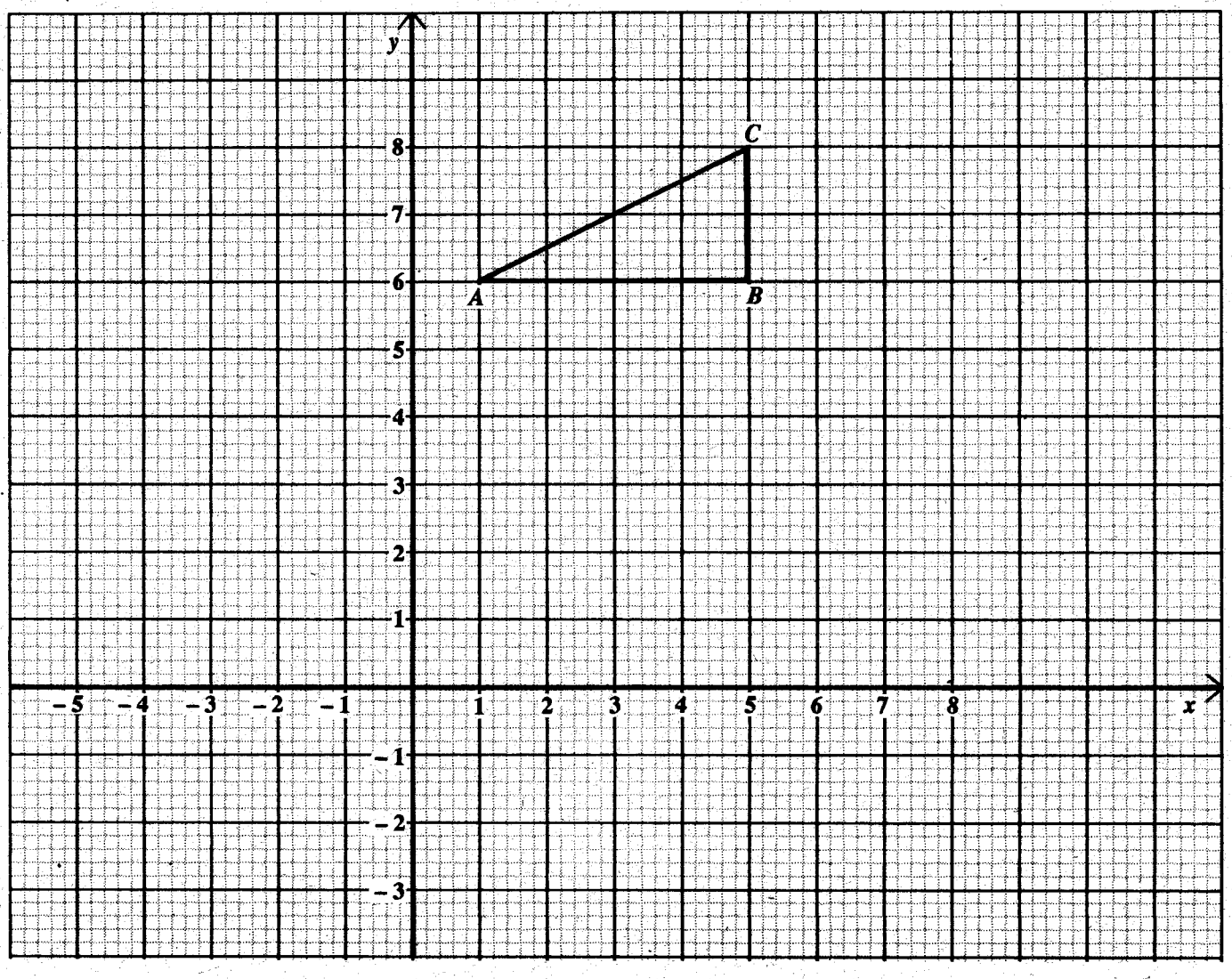
FORM TP 96104

MAY/JUNE 1996

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 – General Proficiency

Answer sheet for Question 5 (b). **Candidate number .....**